1. Organisation of the Organism

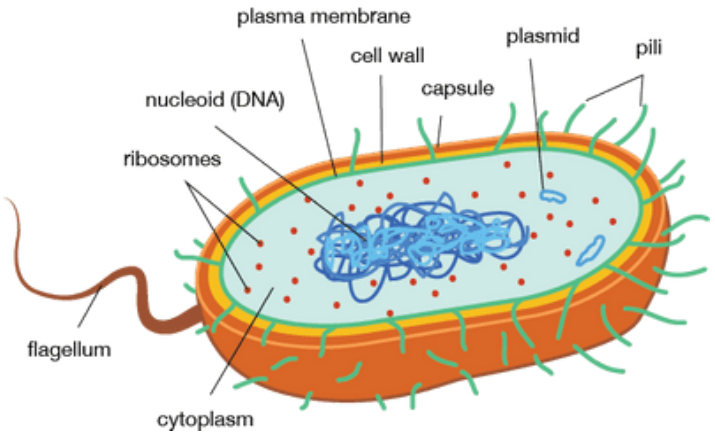
1.1. Cell Structure

- All living things are made of cells.
- New cells are produced by the division of existing cells
- All typical eukaryotic cells (multicellular) have:
  - Cell Membrane: controls movement in and out of cells
  - Cytoplasm: where chemical/metabolic reactions take place
  - Nucleus: contains DNA, stores genetic information and controls the activity of the cell
  - Mitochondria: where aerobic respiration happens
  - Ribosome: allows protein synthesis in the cytoplasm
  - Endoplasmic reticulum: Studded with ribosomes, it looks rough under the microscope.
- A typical animal cell (e.g., the liver cell) has all the above.
- Plant cells especially also have:
  - Vacuole: cell sap to keep cell turgid
  - Cell Wall: rigid to hold the shape of the cell, strengthens the cell
  - Chloroplasts: contain chlorophyll, which absorbs light energy for photosynthesis
- A typical plant cell (e.g., the palisade cell) has everything above.

Prokaryotes

Prokaryotes (Unicellular) DO NOT have mitochondria and a true nucleus.

- One example of a prokaryote is bacteria.
- A bacterial cell only contains a cell wall made of peptidoglycan, cell membrane, cytoplasm, ribosomes, and plasmids.
- It lacks a nucleus and is represented by a circular chromosome of DNA.
- Plasmids are small, circular rings of DNA in the cytoplasm with extra genes outside the chromosomal DNA.



Syllabus 2.1.3: You must be able to identify the cell structures in diagrams and images of plant, animal and bacterial cells

1.4. Levels of Organisation

- The division of existing cells produces new cells.

Key Terms

- Cells: Building Blocks of Life
- Tissue: Groups of cells with similar structures working together to perform a shared function
- Organ: Group of tissues working together to perform a specific function
- Organ system: Group of organs with related functions working together to perform body functions.
- Organism: A human

Specialised Cells

- Specialised Cells have Specific Functions.

Specialised Cells	Specific Function	Location of Cell
Ciliated cells	Movement of mucus in the trachea and bronchi	Respiratory Tract, Fallopian Tube, Testes
Root Hair cells	Absorption	Roots
Palisade Mesophyll cell	Photosynthesis	Leaf Cells
Neurones	Conduction of electrical impulses	Everywhere in an organism
Red Blood cells	Transport of oxygen	Everywhere in an organism
Sperm and Egg cells/ovum (gametes)	For reproduction	In their respective gonads (glands)

1.5. Magnification

The general formula is represented in this ways:

$$Magnification = \frac{\text{size of drawing}}{\text{size of specimen}} = \frac{\text{image}}{\text{actual}} = \frac{I}{A}$$

Other Forms in Magnification Formula

- Actual size = image size ÷ magnification
- Image size = magnification x actual size

Unit Conversions (µm - micrometre)

- 1cm = 10mm
- 1mm = 1000µm
- 1µm = 0.001mm

Magnification does NOT have any units ('x 50' or 'x 5000')

Tip: This comes out frequently in all three papers

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